# Supported Command List - V2.1

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# Version history

| Version | Publish date | Description |
|---|---|---|
| 1.0.0 | 06 Jan 2023 | Add example to do Connect payload<br>Add example to do Load payload settings<br>Add example to do Set payload settings<br>Add example to do Image capture with the payload<br>Add example to do Video record with the payload<br>Add example to do Get video streaming from payload |
| 1.0.1 | 25 May 2023 | Included Gremsy gSDK to communicate and control gimbal<br>Add examples to move gimbal in Angle mode and Angular rate mode using Mavlink Gimbal Protocol v2 |
| v1.1.0 | 30 July 2024 | Add some examples<br>Get LRF status for VIO<br>Payload define |
| v2.0.0 | 16 Aug 2024 | Replace gSDK by MAVLINK interface<br>Provide MAVLINK command list<br>Publish resource |
| v2.1.0 | 04 Nov 2026 | Modify Set gimbal mode command<br>Add set gimbal forward flag command |

# Payload mavlink target

| Encode a command_long struct | Values | Description |
|---|---|---|
| target_system | 1 | |
| target_component | 191 | MAV_COMP_ID_ONBOARD_COMPUTER |
| chan | 0 | MAVLink channel |

# Command mavlink list

**Set camera running mode**

- Message: COMMAND_LONG (76)

| Field Name | Type | Values | Description |
|---|---|---|---|
| target_system | uint8_t | 1 | System ID |
| target_component | uint8_t | 101 (MAV_COMP_ID_CAMERA2) | MAVLINK Camera Component ID on Web setting |
| command | uint16_t | MAV_CMD_SET_CAMERA_MODE (530) | MAV_CMD |
| confirmation | uint8_t | 1 | |
| param2 | float | 0<br>1 | Photo mode<br>Video mode |

**Set camera record source**

- Message: PARAM_EXT_SET (323)

| Field Name | Type | Values | Description |
|---|---|---|---|
| target_system | uint8_t | 1 | System ID |
| target_component | uint8_t | 101 (MAV_COMP_ID_CAMERA2) | MAVLINK Camera Component ID on Web setting |
| param_id | char[16] | "C_V_REC" | |
| param_value | char[128] | "0"<br>"1"<br>"2"<br>"5" | Record both<br>Record EO<br>Record IR<br>Record OSD |
| param_type | uint8_t | 5 (MAV_PARAM_EXT_TYPE_UINT32) | MAV_PARAM_EXT_TYPE |

**Set camera view source**

- Message: PARAM_EXT_SET (323)

| Field Name | Type | Values | Description |
|---|---|---|---|
| target_system | uint8_t | 1 | System ID |
| target_component | uint8_t | 101 (MAV_COMP_ID_CAMERA2) | MAVLINK Camera Component ID on Web setting |
| param_id | char[16] | "C_SOURCE" | |
| param_value | char[128] | "0"<br>"1"<br>"2"<br>"3"<br>"4" | EO/IR<br>Only EO<br>Only IR<br>IR/EO<br>Sync |
| param_type | uint8_t | 5 (MAV_PARAM_EXT_TYPE_UINT32) | MAV_PARAM_EXT_TYPE |

**Set EO camera zoom mode**

- Message: PARAM_EXT_SET (323)

| Field Name | Type | Values | Description |
|---|---|---|---|
| target_system | uint8_t | 1 | System ID |
| target_component | uint8_t | 101 (MAV_COMP_ID_CAMERA2) | MAVLINK Camera Component ID on Web setting |
| param_id | char[16] | "C_V_ZM_MODE" | |
| param_value | char[128] | "0"<br>"2" | Combine<br>Super Resolution |
| param_type | uint8_t | 5 (MAV_PARAM_EXT_TYPE_UINT32) | MAV_PARAM_EXT_TYPE |

**Set EO camera zoom - Super Resolution**

- Message: PARAM_EXT_SET (323)

| Field Name | Type | Values | Description |
|---|---|---|---|
| target_system | uint8_t | 1 | System ID |
| target_component | uint8_t | 101 (MAV_COMP_ID_CAMERA2) | MAVLINK Camera Component ID on Web setting |
| param_id | char[16] | "C_V_ZM_SR_LV" | |
| param_value | char[128] | "0"<br>"1" | 1X<br>2X |

| Field Name | Type | Values | Description |
|---|---|---|---|
| | | "2" | 4X |
| | | "3" | 6X |
| | | "4" | 8X |
| | | "5" | 10X |
| | | "6" | 12X |
| | | "7" | 14X |
| | | "8" | 16X |
| | | "9" | 18X |
| | | "10" | 20X |
| | | "11" | 22X |
| | | "12" | 24X |
| | | "13" | 26X |
| | | "14" | 28X |
| | | "15" | 30X |
| param_type | uint8_t | 5 (MAV_PARAM_EXT_TYPE_UINT32) | MAV_PARAM_EXT_TYPE |

**Set EO camera index zoom - Combine**

- Message: PARAM_EXT_SET (323)

| Field Name | Type | Values | Description |
|---|---|---|---|
| target_system | uint8_t | 1 | System ID |
| target_component | uint8_t | 101 (MAV_COMP_ID_CAMERA2) | MAVLINK Camera Component ID on Web setting |
| param_id | char[16] | | "C_V_ZM_CB_LV" |
| param_value | char[128] | "0"<br>"1"<br>"2"<br>"3"<br>"4"<br>"5"<br>"6" | 1X<br>10X<br>20X<br>40X<br>80X<br>120X<br>240X |
| param_type | uint8_t | 5 (MAV_PARAM_EXT_TYPE_UINT32) | MAV_PARAM_EXT_TYPE |

**Set camera zoom type**

- Message: COMMAND_LONG (76)

| Field Name | Type | Values | Description |
|---|---|---|---|
| target_system | uint8_t | 1 | System ID |

| Field Name | Type | Values | Description |
|---|---|---|---|
| target_component | uint8_t | 101 (MAV_COMP_ID_CAMERA2) | MAVLINK Camera Component ID on Web setting |
| command | uint16_t | MAV_CMD_SET_CAMERA_ZOOM (531) | MAV_CMD |
| confirmation | uint8_t | 1 | |
| param1 | float | 0<br>1<br>2 | ZOOM_TYPE_STEP<br>ZOOM_TYPE_CONTINUOUS<br>ZOOM_TYPE_RANGE |
| param2 | float | -1 for wide, 1 for tele<br>-1 for wide, 1 for tele, 0 to stop zooming<br>A percentage value between 0.0 and 100.0 | Zoom one step increment<br>Continuous zoom up/down until stopped<br>Zoom value as proportion of full camera range |

**Set camera focus type**

- Message: COMMAND_LONG (76)

| Field Name | Type | Values | Description |
|---|---|---|---|
| target_system | uint8_t | 1 | System ID |
| target_component | uint8_t | 101 (MAV_COMP_ID_CAMERA2) | MAVLINK Camera Component ID on Web setting |
| command | uint16_t | MAV_CMD_SET_CAMERA_FOCUS (532) | MAV_CMD |
| confirmation | uint8_t | 1 | |
| param1 | float | 1<br>4 | FOCUS_TYPE_CONTINUOUS<br>FOCUS_TYPE_AUTO |
| param2 | float | -1: focusing in, 1: focusing out, 0: stop | Continuous focus up/down until stopped |

**Set camera IR palette**

- Message: PARAM_EXT_SET (323)

| Field Name | Type | Values | Description |
|---|---|---|---|
| target_system | uint8_t | 1 | System ID |
| target_component | uint8_t | 101 (MAV_COMP_ID_CAMERA2) | MAVLINK Camera Component ID on Web setting |

| Field Name | Type | Values | Description |
|---|---|---|---|
| param_id | char[16] | "C_T_PALETTE" | |
| param_value | char[128] | "0"<br>"1"<br>"2"<br>"3"<br>"4"<br>"5"<br>"6"<br>"7"<br>"8"<br>"9" | F1: WhiteHot ; G1: WhiteHot<br>F1: BlackHot ; G1: Fulgurite<br>F1: Rainbow ; G1: IronRed<br>F1: RainbowHC ; G1: HotIron<br>F1: Ironbow ; G1: Medical<br>F1: Lava ; G1: Arctic<br>F1: Arctic ; G1: Rainbow1<br>F1: Globow ; G1: Rainbow2<br>F1: Gradedfire ; G1: Tint<br>F1: Hottest ; G1: BlackHot |
| param_type | uint8_t | 5 (MAV_PARAM_EXT_TYPE_UINT32) | MAV_PARAM_EXT_TYPE |

**Set payload camera start capture image**

- Message: COMMAND_LONG (76)

| Field Name | Type | Values | Description |
|---|---|---|---|
| target_system | uint8_t | 1 | System ID |
| target_component | uint8_t | 101 (MAV_COMP_ID_CAMERA2) | MAVLINK Camera Component ID on Web setting |
| command | uint16_t | MAV_CMD_IMAGE_START_CAPTURE (2000) | MAV_CMD |
| confirmation | uint8_t | 1 | |
| param2 | int | Default, min: 0 | Desired elapsed time between two consecutive pictures (in seconds) |

**Set payload camera stop capture image**

- Message: COMMAND_LONG (76)

| Field Name | Type | Values | Description |
|---|---|---|---|
| target_system | uint8_t | 1 | System ID |
| target_component | uint8_t | 101 (MAV_COMP_ID_CAMERA2) | MAVLINK Camera Component ID on Web setting |

| Field Name | Type | Values | Description |
|---|---|---|---|
| command | uint16_t | MAV_CMD_IMAGE_STOP_CAPTURE (2001) | MAV_CMD |
| confirmation | uint8_t | 1 | |

**Set payload camera start record video**

- Message: COMMAND_LONG (76)

| Field Name | Type | Values | Description |
|---|---|---|---|
| target_system | uint8_t | 1 | System ID |
| target_component | uint8_t | 101 (MAV_COMP_ID_CAMERA2) | MAVLINK Camera Component ID on Web setting |
| command | uint16_t | MAV_CMD_VIDEO_START_CAPTURE (2500) | MAV_CMD |
| confirmation | uint8_t | 1 | |

**Set payload camera stop record video**

- Message: COMMAND_LONG (76)

| Field Name | Type | Values | Description |
|---|---|---|---|
| target_system | uint8_t | 1 | System ID |
| target_component | uint8_t | 101 (MAV_COMP_ID_CAMERA2) | MAVLINK Camera Component ID on Web setting |
| command | uint16_t | MAV_CMD | MAV_CMD_VIDEO_STOP_CAPTURE (2501) |
| confirmation | uint8_t | 1 | |

**Set camera video flip**

- Message: PARAM_EXT_SET (323)

| Field Name | Type | Values | Description |
|---|---|---|---|
| target_system | uint8_t | 1 | System ID |
| target_component | uint8_t | 101 (MAV_COMP_ID_CAMERA2) | MAVLINK Camera Component ID on Web setting |
| param_id | char[16] | "C_V_FLIP" | |

| Field Name | Type | Values | Description |
|---|---|---|---|
| param_value | char[128] | "2"<br>"3" | On<br>Off |
| param_type | uint8_t | 5<br>(MAV_PARAM_EXT_TYPE_UINT32) | MAV_PARAM_EXT_TYPE |

**Set camera video defog**

- Message: PARAM_EXT_SET (323)

| Field Name | Type | Values | Description |
|---|---|---|---|
| target_system | uint8_t | 1 | System ID |
| target_component | uint8_t | 101 (MAV_COMP_ID_CAMERA2) | MAVLINK Camera Component ID on Web setting |
| param_id | char[16] | "C_V_DEFOG" | |
| param_value | char[128] | "0"<br>"1" | Off<br>On |
| param_type | uint8_t | 5<br>(MAV_PARAM_EXT_TYPE_UINT32) | MAV_PARAM_EXT_TYPE |

**Set camera video defog level**

- Message: PARAM_EXT_SET (323)

| Field Name | Type | Values | Description |
|---|---|---|---|
| target_system | uint8_t | 1 | System ID |
| target_component | uint8_t | 101 (MAV_COMP_ID_CAMERA2) | MAVLINK Camera Component ID on Web setting |
| param_id | char[16] | "C_V_DEFOG_LV" | |
| param_value | char[128] | "0"<br>"1"<br>"2"<br>"3" | Lowest<br>Low<br>Mid<br>High |
| param_type | uint8_t | 5<br>(MAV_PARAM_EXT_TYPE_UINT32) | MAV_PARAM_EXT_TYPE |

**Set camera video auto exposure**

- Message: PARAM_EXT_SET (323)

| Field Name | Type | Values | Description |
|---|---|---|---|
| target_system | uint8_t | 1 | System ID |
| target_component | uint8_t | 101 (MAV_COMP_ID_CAMERA2) | MAVLINK Camera Component ID on Web setting |
| param_id | char[16] | "C_V_AE" | |
| param_value | char[128] | "0"<br>"3"<br>"10"<br>"11"<br>"13" | Auto<br>Manual<br>Shutter<br>Iris<br>Bright |
| param_type | uint8_t | 5 (MAV_PARAM_EXT_TYPE_UINT32) | MAV_PARAM_EXT_TYPE |

**Set camera video shutter speed**

- Message: PARAM_EXT_SET (323)

| Field Name | Type | Values | Description |
|---|---|---|---|
| target_system | uint8_t | 1 | System ID |
| target_component | uint8_t | 101 (MAV_COMP_ID_CAMERA2) | MAVLINK Camera Component ID on Web setting |
| param_id | char[16] | "C_V_SP" | |
| param_value | char[128] | "13"<br>"14"<br>"17"<br>"20"<br>"21"<br>"25"<br>"26"<br>"27"<br>"28"<br>"30" | 1/10<br>1/20<br>1/50<br>1/100<br>1/125<br>1/500<br>1/725<br>1/1000<br>1/1500<br>1/2000 |
| param_type | uint8_t | 5 (MAV_PARAM_EXT_TYPE_UINT32) | MAV_PARAM_EXT_TYPE |

**Set camera video aperture value**

- Message: PARAM_EXT_SET (323)

| Field Name | Type | Values | Description |
|---|---|---|---|
| target_system | uint8_t | 1 | System ID |

| Field Name | Type | Values | Description |
|---|---|---|---|
| target_component | uint8_t | 101 (MAV_COMP_ID_CAMERA2) | MAVLINK Camera Component ID on Web setting |
| param_id | char[16] | "C_V_IrP" | |
| param_value | char[128] | 0 | Aperture value can be set from 0 to 25, step 1 |
| param_type | uint8_t | 5 (MAV_PARAM_EXT_TYPE_UINT32) | MAV_PARAM_EXT_TYPE |

**Set camera video bright value**

- Message: PARAM_EXT_SET (323)

| Field Name | Type | Values | Description |
|---|---|---|---|
| target_system | uint8_t | 1 | System ID |
| target_component | uint8_t | 101 (MAV_COMP_ID_CAMERA2) | MAVLINK Camera Component ID on Web setting |
| param_id | char[16] | "C_V_BrP" | |
| param_value | char[128] | 0 | Bright value can be set from 0 to 41, step 1 |
| param_type | uint8_t | 5 (MAV_PARAM_EXT_TYPE_UINT32) | MAV_PARAM_EXT_TYPE |

**Set camera video white balance**

- Message: PARAM_EXT_SET (323)

| Field Name | Type | Values | Description |
|---|---|---|---|
| target_system | uint8_t | 1 | System ID |
| target_component | uint8_t | 101 (MAV_COMP_ID_CAMERA2) | MAVLINK Camera Component ID on Web setting |
| param_id | char[16] | "C_V_WB" | |
| param_value | char[128] | "0"<br>"1"<br>"2"<br>"3"<br>"4"<br>"5" | Auto<br>Indoor<br>Outdoor<br>One push<br>ATW<br>Manual |

| Field Name | Type | Values | Description |
|---|---|---|---|
| param_type | uint8_t | 5 (MAV_PARAM_EXT_TYPE_UINT32) | MAV_PARAM_EXT_TYPE |

**Set gimbal mode**

Note: For Software Package v307 and later, PARAM_EXT_SET must not be used to configure the gimbal mode. Use MAVLINK_MSG_ID_GIMBAL_DEVICE_SET_ATTITUDE (Message ID: 284) instead.

- Message: MAVLINK_MSG_ID_GIMBAL_DEVICE_SET_ATTITUDE (284)

| Field Name | Type | | Values | Description |
|---|---|---|---|---|
| target_system | uint8_t | | 1 | System ID |
| target_component | uint8_t | | 154 (MAV_COMP_ID_GIMBAL) | Gimbal Components ID |
| flags | uint16_t | | 1, 2, 4, 8, 16, 32, 64, 128, 256, 512 | Bitmask for configuring Gimbal modes. Multiple flags can be combined using bitwise OR |
| q | float[4] | rad/s | NaN | Quaternion components, w, x, y, z |
| angular_velocity_x | float | rad/s | 0 | X component of angular velocity (Roll) |
| angular_velocity_y | float | rad/s | 0 | Y component of angular velocity (Pitch) |
| angular_velocity_z | float | rad/s | 0 | Z component of angular velocity (Yaw) |

**Set gimbal forward flag**

- Message: PARAM_EXT_SET (323)

| Field Name | Type | Values | Description |
|---|---|---|---|
| target_system | uint8_t | 1 | System ID |
| target_component | uint8_t | 101 (MAV_COMP_ID_CAMERA2) | MAVLINK Camera Component ID on Web setting |
| param_id | char[16] | "GB_FW_FLAG" | |
| param_value | char[128] | "0"<br>"1"<br>"2" | Autopilot<br>Onboard computer<br>SBUS Convert |

| Field Name | Type | Values | Description |
|---|---|---|---|
| param_type | uint8_t | 5 (MAV_PARAM_EXT_TYPE_UINT32) | MAV_PARAM_EXT_TYPE |

**Set gimbal RC mode**

- Message: PARAM_EXT_SET (323)

| Field Name | Type | Values | Description |
|---|---|---|---|
| target_system | uint8_t | 1 | System ID |
| target_component | uint8_t | 101 (MAV_COMP_ID_CAMERA2) | MAVLINK Camera Component ID on Web setting |
| param_id | char[16] | "RC_MODE" | |
| param_value | char[128] | "0"<br>"1" | Gremsy<br>Standard |
| param_type | uint8_t | 5 (MAV_PARAM_EXT_TYPE_UINT32) | MAV_PARAM_EXT_TYPE |

**Set gimbal move to angle**

- Message: GIMBAL_DEVICE_SET_ATTITUDE (284)

| Field Name | Type | | Values | Description |
|---|---|---|---|---|
| target_system | uint8_t | | 1 | System ID |
| target_component | uint8_t | | 101 (MAV_COMP_ID_CAMERA2) | MAVLINK Camera Component ID on Web setting |
| q | float[4] | rad/s | mavlink_euler_to_quaternion(roll, pitch, yaw, q) | Quaternion components, w, x, y, z |
| angular_velocity_x | float | rad/s | NaN | X component of angular velocity (Roll) |
| angular_velocity_y | float | rad/s | NaN | Y component of angular velocity (Pitch) |
| angular_velocity_z | float | rad/s | NaN | Z component of angular velocity (Yaw) |

**Set gimbal move by speed**

- Message: GIMBAL_DEVICE_SET_ATTITUDE (284)

| Field Name | Type | | Values | Description |
|---|---|---|---|---|
| target_system | uint8_t | | 1 | System ID |
| target_component | uint8_t | | 101 (MAV_COMP_ID_CAMERA2) | MAVLINK Camera Component ID on Web setting |
| q | float[4] | rad/s | NaN | Quaternion components, w, x, y, z |
| angular_velocity_x | float | rad/s | Speed roll | X component of angular velocity (Roll) |
| angular_velocity_y | float | rad/s | Speed pitch | Y component of angular velocity (Pitch) |
| angular_velocity_z | float | rad/s | Speed yaw | Z component of angular velocity (Yaw) |

**Do object tracking**

- Message: COMMAND_LONG (76)

| Field Name | Type | Values | Description |
|---|---|---|---|
| target_system | uint8_t | 1 | System ID |
| target_component | uint8_t | 26 (MAV_COMP_ID_USER2) | MAVLINK Camera Component ID on Web setting |
| command | uint16_t | 31013 (MAV_CMD_USER_4) | Command ID (of command to send) |
| confirmation | uint8_t | 1 | |
| param1 | float | 1 | |
| param2 | float | 1 | |
| param3 | float | Do not use | |
| param4 | float | 0<br>1 | Traking cancel<br>Traking active |
| param5 | float | Do not use | |
| param6 | float | x | Center of the tracking box on the x-axis |
| param7 | float | y | Center of the tracking box on the y-axis |

**Do object detection**

- Message: PARAM_EXT_SET (323)

| Field Name | Type | Values | Description |
|---|---|---|---|
| target_system | uint8_t | 1 | System ID |
| target_component | uint8_t | 101 (MAV_COMP_ID_CAMERA2) | MAVLINK Camera Component ID on Web setting |
| param_id | char[16] | "TRACK_MODE" | |
| param_value | char[128] | "0"<br>"1" | Disable<br>Enable |
| param_type | uint8_t | 5<br>(MAV_PARAM_EXT_TYPE_UINT32) | MAV_PARAM_EXT_TYPE |

**Get payload camera information**

- Message: COMMAND_LONG (76)

| Field Name | Type | Values | Description |
|---|---|---|---|
| target_system | uint8_t | 1 | System ID |
| target_component | uint8_t | 101 (MAV_COMP_ID_CAMERA2) | MAVLINK Camera Component ID on Web setting |
| command | uint16_t | MAV_CMD_REQUEST_CAMERA_INFORMATION (521) | MAV_CMD |
| confirmation | uint8_t | 1 | |

**Get payload camera streaming information**

- Message: COMMAND_LONG (76)

| Field Name | Type | Values | Description |
|---|---|---|---|
| target_system | uint8_t | 1 | System ID |
| target_component | uint8_t | 101 (MAV_COMP_ID_CAMERA2) | MAVLINK Camera Component ID on Web setting |
| command | uint16_t | MAV_CMD_REQUEST_VIDEO_STREAM_INFORMATION (2504) | MAV_CMD |
| confirmation | uint8_t | 1 | |

**Get payload storage**

- Message: COMMAND_LONG (76)

| Field Name | Type | Values | Description |
|---|---|---|---|
| target_system | uint8_t | 1 | System ID |
| target_component | uint8_t | 101 (MAV_COMP_ID_CAMERA2) | MAVLINK Camera Component ID on Web setting |
| command | uint16_t | MAV_CMD_REQUEST_STORAGE_INFORMATION (525) | MAV_CMD |
| confirmation | uint8_t | 1 | |

**Get payload capture status**

- Message: COMMAND_LONG (76)

| Field Name | Type | Values | Description |
|---|---|---|---|
| target_system | uint8_t | 1 | System ID |
| target_component | uint8_t | 101 (MAV_COMP_ID_CAMERA2) | MAVLINK Camera Component ID on Web setting |
| command | uint16_t | MAV_CMD_REQUEST_CAMERA_CAPTURE_STATUS (527) | MAV_CMD |
| confirmation | uint8_t | 1 | |

**Get payload camera mode**

- Message: COMMAND_LONG (76)

| Field Name | Type | Values | Description |
|---|---|---|---|
| target_system | uint8_t | 1 | System ID |
| target_component | uint8_t | 101 (MAV_COMP_ID_CAMERA2) | MAVLINK Camera Component ID on Web setting |
| command | uint16_t | MAV_CMD_REQUEST_CAMERA_SETTINGS (522) | MAV_CMD |
| confirmation | uint8_t | 1 | |